

MICROPROCESSOR 8085

COMPLETE LAB REPORT

All Laboratory Tasks - Programs, Algorithms, Memory Organization, Hex Codes and Verified Outputs

Name	_____
Roll No.	_____
Program / Semester	_____
College	_____
Submitted To	_____

Prepared in the same lab-report structure as the provided sample.

Report Conventions

The following conventions are used consistently throughout the report:

- Program execution address is assumed to begin at 2500H.
- Input data is generally kept from 2000H onward and results from 3000H onward.
- For a 16-bit value, the lower byte (LSB) is stored at the lower memory address and the higher byte (MSB) at the next address, matching 8085 LHLD/SHLD behavior.
- All values are hexadecimal unless explicitly stated otherwise.
- The Hexcode Table gives the exact machine bytes in little-endian address order and the instruction memory size.
- The outputs shown were independently verified against the listed instruction bytes and sample data.

Submission Format

As in the supplied example, the Algorithm and Program sections can be handwritten in the practical file. The Hexcode Table and Output section can be used directly while entering and checking the program in an 8085 simulator.

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Lab Task 1A: Load an 8-bit Data into a Register

Question

Write an 8085 program to load an 8-bit data from memory into a register.

Solution

Algorithm

1. Start the program.
2. Load the 8-bit data from memory location 2000H into the accumulator.
3. Transfer the accumulator content to register B.
4. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	45H	Input data
Register B	45H	Final register value

Program

```

LDA 2000H      ; Load the data at 2000H into A
MOV B, A       ; Copy A into register B
HLT           ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LDA 2000H	3A 00 20	3 Bytes
2503H	MOV B, A	47	1 Byte
2504H	HLT	76	1 Byte

Output

```

Input: [2000H] = 45H
Output: B = 45H

```

Lab Task 1B: Transfer Data Between Registers

Question

Write an 8085 program to transfer data between registers.

Solution

Algorithm

1. Start the program.
2. Load 32H into register B.
3. Transfer the content of B to register C.
4. Stop the program.

Memory Organization

Location / Register	Value	Purpose
Register B	32H	Source register
Register C	32H	Destination register after transfer

Program

```

MVI B, 32H      ; Load 32H into B
MOV C, B        ; Copy B into C
HLT             ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	MVI B, 32H	06 32	2 Bytes
2502H	MOV C, B	48	1 Byte
2503H	HLT	76	1 Byte

Output

```

Output: B = 32H
Output: C = 32H

```

Lab Task 1C: Transfer a Specific Data Byte to a Register

Question

Write an 8085 program to transfer a specific 8-bit data byte directly to a register.

Solution

Algorithm

1. Start the program.
2. Load the immediate data byte 5AH into register C.
3. Stop the program.

Memory Organization

Location / Register	Value	Purpose
Register C	5AH	Final register value

Program

```
MVI C, 5AH      ; Load immediate 5AH into C
HLT            ; Stop the program
```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	MVI C, 5AH	0E 5A	2 Bytes
2502H	HLT	76	1 Byte

Output

Output: C = 5AH

Lab Task 1D: Transfer a Specific Data Byte to a Memory Location

Question

Write an 8085 program to transfer a specific 8-bit data byte to a memory location.

Solution

Algorithm

1. Start the program.
2. Load the immediate data byte 6AH into the accumulator.
3. Store the accumulator content at memory location 3000H.
4. Stop the program.

Memory Organization

Location / Register	Value	Purpose
3000H	6AH	Stored data byte

Program

```

MVI A, 6AH      ; Load immediate 6AH into A
STA 3000H      ; Store A at 3000H
HLT           ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	MVI A, 6AH	3E 6A	2 Bytes
2502H	STA 3000H	32 00 30	3 Bytes
2505H	HLT	76	1 Byte

Output

Output: [3000H] = 6AH

Lab Task 1E: Transfer Data from Memory to a Register

Question

Write an 8085 program to transfer an 8-bit data from a memory location to a register.

Solution

Algorithm

1. Start the program.
2. Load the data from memory location 2000H into the accumulator.
3. Transfer the accumulator content to register D.
4. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	7CH	Input data
Register D	7CH	Final register value

Program

```

LDA 2000H      ; Load [2000H] into A
MOV D, A       ; Copy A into D
HLT           ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LDA 2000H	3A 00 20	3 Bytes
2503H	MOV D, A	57	1 Byte
2504H	HLT	76	1 Byte

Output

```

Input: [2000H] = 7CH
Output: D = 7CH

```

Lab Task 2: Addition of Two Hexadecimal Numbers and Output to PORT1

Question

Write an 8085 program to add two hexadecimal numbers and display the sum at output PORT1.

Solution

Algorithm

1. Start the program.
2. Load the first hexadecimal number 25H into the accumulator.
3. Load the second hexadecimal number 37H into register B.
4. Add register B to the accumulator.
5. Send the sum to output port 01H (PORT1).
6. Stop the program.

Memory Organization

Location / Register	Value	Purpose
PORT 01H	5CH	Sum displayed at PORT1

Program

```

MVI A, 25H      ; Load first number into A
MVI B, 37H      ; Load second number into B
ADD B           ; A <- A + B
OUT 01H         ; Send A to PORT1
HLT            ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	MVI A, 25H	3E 25	2 Bytes
2502H	MVI B, 37H	06 37	2 Bytes
2504H	ADD B	80	1 Byte
2505H	OUT 01H	D3 01	2 Bytes
2507H	HLT	76	1 Byte

Output

```

25H + 37H = 5CH
Output at PORT 01H = 5CH

```


Lab Task 3: Subtraction of Two 8-bit Hexadecimal Numbers

Question

Write an 8085 program to subtract two 8-bit hexadecimal numbers.

Solution

Algorithm

1. Start the program.
2. Load 5AH into the accumulator as the minuend.
3. Load 27H into register B as the subtrahend.
4. Subtract B from the accumulator.
5. Store the result at memory location 3000H.
6. Stop the program.

Memory Organization

Location / Register	Value	Purpose
3000H	33H	8-bit subtraction result

Program

```

MVI A, 5AH      ; Load minuend into A
MVI B, 27H      ; Load subtrahend into B
SUB B           ; A ← A - B
STA 3000H       ; Store result at 3000H
HLT            ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	MVI A, 5AH	3E 5A	2 Bytes
2502H	MVI B, 27H	06 27	2 Bytes
2504H	SUB B	90	1 Byte
2505H	STA 3000H	32 00 30	3 Bytes
2508H	HLT	76	1 Byte

Output

```

5AH - 27H = 33H
Output: [3000H] = 33H

```

Lab Task 4: Addition of Two 16-bit Hexadecimal Numbers

Question

Write an 8085 program to add two 16-bit hexadecimal numbers stored from 2000H to 2003H and store the 16-bit result at 3000H and 3001H.

Solution

Algorithm

1. Start the program.
2. Load the first 16-bit number from 2000H-2001H into the HL register pair.
3. Exchange HL with DE so that the first number is preserved in DE.
4. Load the second 16-bit number from 2002H-2003H into HL.
5. Add DE to HL using DAD D.
6. Store the 16-bit result from HL at 3000H-3001H.
7. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	34H	First number LSB
2001H	12H	First number MSB
2002H	11H	Second number LSB
2003H	11H	Second number MSB
3000H	45H	Result LSB
3001H	23H	Result MSB

Program

```

LHLD 2000H    ; Load first 16-bit number into HL
XCHG         ; Transfer first number from HL to DE
LHLD 2002H    ; Load second 16-bit number into HL
DAD D         ; HL <- HL + DE
SHLD 3000H    ; Store result at 3000H and 3001H
HLT          ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LHLD 2000H	2A 00 20	3 Bytes
2503H	XCHG	EB	1 Byte
2504H	LHLD 2002H	2A 02 20	3 Bytes
2507H	DAD D	19	1 Byte
2508H	SHLD 3000H	22 00 30	3 Bytes
250BH	HLT	76	1 Byte

Output

```

First number = 1234H
Second number = 1111H
Result = 2345H
[3000H] = 45H (LSB), [3001H] = 23H (MSB)

```

Lab Task 5: Subtraction of Two 16-bit Hexadecimal Numbers

Question

Write an 8085 program to subtract one 16-bit hexadecimal number from another 16-bit hexadecimal number.

Solution

Algorithm

1. Start the program.
2. Load the first 16-bit number from 2000H-2001H into HL and exchange it into DE.
3. Load the second 16-bit number from 2002H-2003H into HL.
4. Subtract the lower byte L from E and place the result in L.
5. Subtract the higher byte H from D with borrow using SBB H and place the result in H.
6. Store the 16-bit result from HL at 3000H-3001H.
7. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	67H	First number LSB
2001H	45H	First number MSB
2002H	34H	Second number LSB
2003H	12H	Second number MSB
3000H	33H	Result LSB
3001H	33H	Result MSB

Program

```

LHLD 2000H    ; Load first number into HL
XCHG          ; Move first number to DE
LHLD 2002H    ; Load second number into HL
MOV A, E       ; Move first-number LSB to A
SUB L          ; Subtract second-number LSB
MOV L, A       ; Save lower result in L
MOV A, D       ; Move first-number MSB to A
SBB H          ; Subtract MSB including borrow
MOV H, A       ; Save higher result in H
SHLD 3000H    ; Store 16-bit result
HLT           ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LHLD 2000H	2A 00 20	3 Bytes
2503H	XCHG	EB	1 Byte
2504H	LHLD 2002H	2A 02 20	3 Bytes
2507H	MOV A, E	7B	1 Byte
2508H	SUB L	95	1 Byte
2509H	MOV L, A	6F	1 Byte
250AH	MOV A, D	7A	1 Byte
250BH	SBB H	9C	1 Byte
250CH	MOV H, A	67	1 Byte
250DH	SHLD 3000H	22 00 30	3 Bytes
2510H	HLT	76	1 Byte

Output

```

4567H - 1234H = 3333H
[3000H] = 33H (LSB), [3001H] = 33H (MSB)

```

Lab Task 6: Multiplication of Two 8-bit Hexadecimal Numbers

Question

Write an 8085 program to multiply two 8-bit hexadecimal numbers.

Solution

Algorithm

1. Start the program.
2. Load the multiplicand from 2000H into E and clear D, forming the 16-bit value DE.
3. Load the multiplier from 2001H into register C.
4. Clear HL to 0000H for the 16-bit product.
5. If the multiplier is zero, store 0000H as the product.
6. Otherwise repeatedly add DE to HL and decrement C until C becomes zero.
7. Store the 16-bit product at 3000H-3001H and stop.

Memory Organization

Location / Register	Value	Purpose
2000H	25H	Multiplicand
2001H	12H	Multiplier
3000H	9AH	Product LSB
3001H	02H	Product MSB

Program

```

LDA 2000H    ; Load multiplicand into A
MOV E, A     ; Move multiplicand to E
MVI D, 00H   ; Clear D to form DE
LDA 2001H    ; Load multiplier into A
MOV C, A     ; Move multiplier to C
LXI H, 0000H ; Clear HL for product
MOV A, C     ; Copy multiplier to A for zero test
ORA A        ; Set flags according to multiplier
JZ 2517H     ; If multiplier is zero, go to STORE
DAD D        ; LOOP: HL <- HL + DE
DCR C        ; Decrement multiplier count
JNZ 2512H    ; Repeat until C = 00H
SHLD 3000H   ; STORE: save 16-bit product
HLT          ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LDA 2000H	3A 00 20	3 Bytes
2503H	MOV E, A	5F	1 Byte
2504H	MVI D, 00H	16 00	2 Bytes
2506H	LDA 2001H	3A 01 20	3 Bytes
2509H	MOV C, A	4F	1 Byte
250AH	LXI H, 0000H	21 00 00	3 Bytes
250DH	MOV A, C	79	1 Byte
250EH	ORA A	B7	1 Byte
250FH	JZ 2517H	CA 17 25	3 Bytes
2512H	DAD D	19	1 Byte
2513H	DCR C	0D	1 Byte
2514H	JNZ 2512H	C2 12 25	3 Bytes
2517H	SHLD 3000H	22 00 30	3 Bytes
251AH	HLT	76	1 Byte

Output

```

25H x 12H = 029AH
[3000H] = 9AH (LSB), [3001H] = 02H (MSB)

```

Lab Task 7: Division of Two 8-bit Hexadecimal Numbers

Question

Write an 8085 program to perform division using two 8-bit hexadecimal numbers and store both quotient and remainder.

Solution

Algorithm

1. Start the program.
2. Load the dividend from 2000H into register B.
3. Load the non-zero divisor from 2001H into register C.
4. Clear register D to hold the quotient and copy the dividend to A.
5. Compare A with the divisor. If A is smaller, the division is complete.
6. Otherwise subtract the divisor from A, increment the quotient, and repeat.
7. Store the remainder at 3001H and the quotient at 3000H.
8. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	07H	Dividend
2001H	05H	Divisor (must be non-zero)
3000H	01H	Quotient
3001H	02H	Remainder

Program

```

LDA 2000H    ; Load dividend
MOV B, A     ; Save dividend in B
LDA 2001H    ; Load divisor
MOV C, A     ; Save divisor in C
MVI D, 00H   ; Clear quotient
MOV A, B     ; Load current remainder into A
CMP C        ; LOOP: compare remainder with divisor
JC 2514H     ; If remainder < divisor, finish
SUB C        ; Subtract divisor
INR D        ; Increment quotient
JMP 250BH    ; Repeat comparison
STA 3001H    ; DONE: store remainder
MOV A, D     ; Move quotient to A
STA 3000H    ; Store quotient
HLT          ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LDA 2000H	3A 00 20	3 Bytes
2503H	MOV B, A	47	1 Byte
2504H	LDA 2001H	3A 01 20	3 Bytes
2507H	MOV C, A	4F	1 Byte
2508H	MVI D, 00H	16 00	2 Bytes
250AH	MOV A, B	78	1 Byte
250BH	CMP C	B9	1 Byte
250CH	JC 2514H	DA 14 25	3 Bytes
250FH	SUB C	91	1 Byte
2510H	INR D	14	1 Byte
2511H	JMP 250BH	C3 0B 25	3 Bytes
2514H	STA 3001H	32 01 30	3 Bytes
2517H	MOV A, D	7A	1 Byte
2518H	STA 3000H	32 00 30	3 Bytes
251BH	HLT	76	1 Byte

Output

```

07H / 05H = quotient 01H, remainder 02H
[3000H] = 01H, [3001H] = 02H

```

Lab Task 8: Check Whether an 8-bit Data is Even or Odd

Question

Write an 8085 program to check whether a specified 8-bit data is even or odd.

Solution

Algorithm

1. Start the program.
2. Load the data from memory location 2000H into the accumulator.
3. Mask all bits except the least significant bit using ANI 01H.
4. If the result is zero, the number is even; store 00H at 3000H.
5. Otherwise the number is odd; store 01H at 3000H.
6. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	36H	Sample input data
3000H	00H	Status: 00H = even, 01H = odd

Program

```

LDA 2000H    ; Load input data
ANI 01H      ; Test the least significant bit
JZ 250EH     ; Jump to EVEN when bit 0 = 0
MVI A, 01H   ; Odd status code
STA 3000H    ; Store odd status
HLT          ; Stop after odd case
MVI A, 00H   ; EVEN: even status code
STA 3000H    ; Store even status
HLT          ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LDA 2000H	3A 00 20	3 Bytes
2503H	ANI 01H	E6 01	2 Bytes
2505H	JZ 250EH	CA 0E 25	3 Bytes
2508H	MVI A, 01H	3E 01	2 Bytes
250AH	STA 3000H	32 00 30	3 Bytes
250DH	HLT	76	1 Byte
250EH	MVI A, 00H	3E 00	2 Bytes
2510H	STA 3000H	32 00 30	3 Bytes
2513H	HLT	76	1 Byte

Output

```

For input 36H: [3000H] = 00H -> EVEN
For input 35H: [3000H] = 01H -> ODD

```

Lab Task 9: Find the Largest Number in an Array

Question

Write an 8085 program to find the largest number in an array of 8-bit data.

Solution

Algorithm

1. Start the program.
2. Store the number of array elements at 2000H and the elements from 2001H onward.
3. Load the count into C and the first array element into A as the current largest value.
4. For each remaining element, compare A with the current memory element.
5. If A is smaller, replace A with the current element; otherwise retain A.
6. Repeat until all elements have been checked.
7. Store the largest value at 3000H and stop.

Memory Organization

Location / Register	Value	Purpose
2000H	05H	Number of elements
2001H	25H	Element 1
2002H	7AH	Element 2
2003H	33H	Element 3
2004H	9CH	Element 4
2005H	41H	Element 5
3000H	9CH	Largest element

Program

```

LXI H, 2000H    ; Point HL to array count
MOV C, M        ; Load count into C
INX H           ; Point to first element
MOV A, M        ; Use first element as current largest
DCR C           ; One element already processed
JZ 2514H        ; If count was 1, store result
INX H           ; LOOP: point to next element
CMP M           ; Compare current largest with element
JNC 2510H       ; If A >= M, keep A
MOV A, M        ; Otherwise update largest
DCR C           ; Decrease remaining count
JNZ 250AH       ; Repeat while elements remain
STA 3000H       ; STORE: save largest value
HLT             ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LXI H, 2000H	21 00 20	3 Bytes
2503H	MOV C, M	4E	1 Byte
2504H	INX H	23	1 Byte
2505H	MOV A, M	7E	1 Byte
2506H	DCR C	0D	1 Byte
2507H	JZ 2514H	CA 14 25	3 Bytes
250AH	INX H	23	1 Byte
250BH	CMP M	BE	1 Byte
250CH	JNC 2510H	D2 10 25	3 Bytes
250FH	MOV A, M	7E	1 Byte
2510H	DCR C	0D	1 Byte
2511H	JNZ 250AH	C2 0A 25	3 Bytes
2514H	STA 3000H	32 00 30	3 Bytes
2517H	HLT	76	1 Byte

Output

```

Array = 25H, 7AH, 33H, 9CH, 41H
Largest value = 9CH
Output: [3000H] = 9CH

```

Lab Task 10: Find the Smallest Number in an Array

Question

Write an 8085 program to find the smallest number in an array of 8-bit data.

Solution

Algorithm

1. Start the program.
2. Store the number of array elements at 2000H and the elements from 2001H onward.
3. Load the count into C and the first array element into A as the current smallest value.
4. For each remaining element, compare A with the current memory element.
5. If A is greater than or equal to the current element, update A with that element; if A is smaller, retain A.
6. Repeat until all elements have been checked.
7. Store the smallest value at 3000H and stop.

Memory Organization

Location / Register	Value	Purpose
2000H	05H	Number of elements
2001H	25H	Element 1
2002H	7AH	Element 2
2003H	33H	Element 3
2004H	09H	Element 4
2005H	41H	Element 5
3000H	09H	Smallest element

Program

```

LXI H, 2000H    ; Point HL to array count
MOV C, M        ; Load count into C
INX H           ; Point to first element
MOV A, M        ; Use first element as current smallest
DCR C           ; One element already processed
JZ 2514H        ; If count was 1, store result
INX H           ; LOOP: point to next element
CMP M           ; Compare current smallest with element
JC 2510H        ; If A < M, keep A
MOV A, M        ; Otherwise update smallest
DCR C           ; Decrease remaining count
JNZ 250AH       ; Repeat while elements remain
STA 3000H       ; STORE: save smallest value
HLT             ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LXI H, 2000H	21 00 20	3 Bytes
2503H	MOV C, M	4E	1 Byte
2504H	INX H	23	1 Byte
2505H	MOV A, M	7E	1 Byte
2506H	DCR C	0D	1 Byte
2507H	JZ 2514H	CA 14 25	3 Bytes
250AH	INX H	23	1 Byte
250BH	CMP M	BE	1 Byte
250CH	JC 2510H	DA 10 25	3 Bytes
250FH	MOV A, M	7E	1 Byte
2510H	DCR C	0D	1 Byte
2511H	JNZ 250AH	C2 0A 25	3 Bytes
2514H	STA 3000H	32 00 30	3 Bytes
2517H	HLT	76	1 Byte

Output

```

Array = 25H, 7AH, 33H, 09H, 41H
Smallest value = 09H
Output: [3000H] = 09H

```


Lab Task 11: Fibonacci Series up to Seven Terms

Question

Write an 8085 program to generate the Fibonacci series up to seven terms.

Solution

Algorithm

1. Start the program.
2. Point HL to 3000H, the start of the output area.
3. Store the first two terms 00H and 01H at 3000H and 3001H.
4. Set D = 05H because five more terms are required.
5. Add the previous two terms to generate the next term and store it through HL.
6. Shift the two working terms forward, decrement D, and repeat until five new terms are generated.
7. Stop the program.

Memory Organization

Location / Register	Value	Purpose
3000H	00H	1st term
3001H	01H	2nd term
3002H	01H	3rd term
3003H	02H	4th term
3004H	03H	5th term
3005H	05H	6th term
3006H	08H	7th term

Program

```

LXI H, 3000H    ; Point HL to output area
MVI B, 00H      ; First Fibonacci term
MOV M, B        ; Store first term
INX H           ; Point to next location
MVI C, 01H      ; Second Fibonacci term
MOV M, C        ; Store second term
INX H           ; Point to next output location
MVI D, 05H      ; Five more terms to generate
MOV A, B        ; LOOP: A <- previous term
ADD C           ; A <- previous + current
MOV M, A        ; Store next term
INX H           ; Advance output pointer
MOV B, C        ; Shift current term into B
MOV C, A        ; Shift new term into C
DCR D           ; Decrease term counter
JNZ 250DH       ; Repeat until seven terms are stored
HLT             ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LXI H, 3000H	21 00 30	3 Bytes
2503H	MVI B, 00H	06 00	2 Bytes
2505H	MOV M, B	70	1 Byte
2506H	INX H	23	1 Byte
2507H	MVI C, 01H	0E 01	2 Bytes
2509H	MOV M, C	71	1 Byte
250AH	INX H	23	1 Byte
250BH	MVI D, 05H	16 05	2 Bytes
250DH	MOV A, B	78	1 Byte
250EH	ADD C	81	1 Byte
250FH	MOV M, A	77	1 Byte
2510H	INX H	23	1 Byte
2511H	MOV B, C	41	1 Byte
2512H	MOV C, A	4F	1 Byte
2513H	DCR D	15	1 Byte
2514H	JNZ 250DH	C2 0D 25	3 Bytes
2517H	HLT	76	1 Byte

Output

Fibonacci series stored at 3000H-3006H = 00H, 01H, 01H, 02H, 03H, 05H, 08H

Lab Task 12: Find the Sum of the Two Bytes of a 16-bit Number

Question

Write an 8085 program to find the sum of a 16-bit number. (Implemented as the sum of its lower and higher 8-bit bytes.)

Solution

Algorithm

1. Start the program.
2. Load the 16-bit number from 2000H-2001H into HL.
3. Move the lower byte L to the accumulator.
4. Add the higher byte H to the accumulator.
5. Store the low byte of the sum at 3000H.
6. Convert the carry, if any, into 00H or 01H and store it at 3001H.
7. Stop the program.

Memory Organization

Location / Register	Value	Purpose
2000H	34H	16-bit number LSB
2001H	12H	16-bit number MSB
3000H	46H	Sum LSB
3001H	00H	Carry / sum MSB

Program

```

LHLD 2000H    ; Load 16-bit number into HL
MOV A, L      ; Move lower byte to A
ADD H         ; Add higher byte
STA 3000H     ; Store low byte of sum
MVI A, 00H    ; Clear A without changing CY
ACI 00H       ; A <- 00H + carry
STA 3001H     ; Store carry as higher byte
HLT          ; Stop the program

```

Hexcode Table

Address	Mnemonic	Hex Code	Memory Size
2500H	LHLD 2000H	2A 00 20	3 Bytes
2503H	MOV A, L	7D	1 Byte
2504H	ADD H	84	1 Byte
2505H	STA 3000H	32 00 30	3 Bytes
2508H	MVI A, 00H	3E 00	2 Bytes
250AH	ACI 00H	CE 00	2 Bytes
250CH	STA 3001H	32 01 30	3 Bytes
250FH	HLT	76	1 Byte

Output

```

16-bit number = 1234H
12H + 34H = 0046H
[3000H] = 46H, [3001H] = 00H

```

Interpretation note: The lab sheet wording is ambiguous. Task 4 already covers addition of two 16-bit numbers, so this task is interpreted as adding the two bytes of one 16-bit number. If your instructor intended two-number 16-bit addition, Task 4 is the exact required program.

Final Verification Checklist

- All listed instruction opcodes and byte counts have been checked.
- All branch destination addresses correspond to a program start address of 2500H.
- 16-bit memory organization uses LSB at the lower address and MSB at the next address.
- The sample numerical outputs are consistent with the stated inputs.
- For Task 7, the divisor must be non-zero.
- For Tasks 9 and 10, the element count at 2000H must be at least 01H.

This report is ready to copy into a handwritten lab file and to use for simulator entry and output checking.